

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
iris = load_iris()
```

```
df = pd.DataFrame(
    iris.data,
    columns=iris.feature_names
)
```

```
df["species"] = [
    iris.target_names[i]
    for i in iris.target
]
```

```
sns.set_style("whitegrid")
```

```
print("Dataset Shape:", df.shape)
```

```
print("\nColumns:")
print(df.columns)
```

```
print("\nFirst 5 Rows:")
print(df.head())
```

```
num_cols = df.select_dtypes(include=['int64', 'float64']).columns
for col in num_cols:
    plt.figure(figsize=(6,4))
    sns.histplot(df[col], kde=True, color="skyblue")
    plt.title(f"Histogram of {col}")
    plt.xlabel(col)
    plt.ylabel("Frequency")
    plt.show()
for col in num_cols:
    plt.figure(figsize=(6,4))
    sns.boxplot(x=df[col], color="orange")
    plt.title(f"Boxplot of {col}")
    plt.show()

plt.figure(figsize=(8,6))
sns.heatmap(
    df[num_cols].corr(),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Heatmap")
plt.show()

sns.pairplot(df, hue="species")
plt.show()

plt.figure(figsize=(6,4))

sns.scatterplot(
    data=df,
```

```
    x=num_cols[0],
    y=num_cols[1],
    hue="species"
)

plt.title(f"{num_cols[0]} vs {num_cols[1]}")
plt.show()
plt.figure(figsize=(6,4))

df["species"].value_counts().plot(
    kind="bar",
    color=["red", "green", "blue"]
)

plt.title("Species Count")
plt.xlabel("Species")
plt.ylabel("Count")
plt.show()
plt.figure(figsize=(6,6))

df["species"].value_counts().plot(
    kind="pie",
    autopct="%1.1f%%"
)

plt.title("Species Distribution")
plt.ylabel("")
plt.show()
plt.figure(figsize=(8,4))

plt.plot(df[num_cols[0]], color="purple")
```

```
plt.title(f"Trend of {num_cols[0]}")  
plt.xlabel("Index")  
plt.ylabel(num_cols[0])  
plt.show()
```

```
print("\nData Visualization Completed Successfully!")  
print("Charts Generated:")  
print("✓ Histogram")  
print("✓ Box Plot")  
print("✓ Correlation Heatmap")  
print("✓ Pair Plot")  
print("✓ Scatter Plot")  
print("✓ Bar Chart")  
print("✓ Pie Chart")  
print("✓ Line Chart")
```

Dataset Shape: (150, 5)

Columns:

```
Index(['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)',  
      'petal width (cm)', 'species'],  
      dtype='object')
```

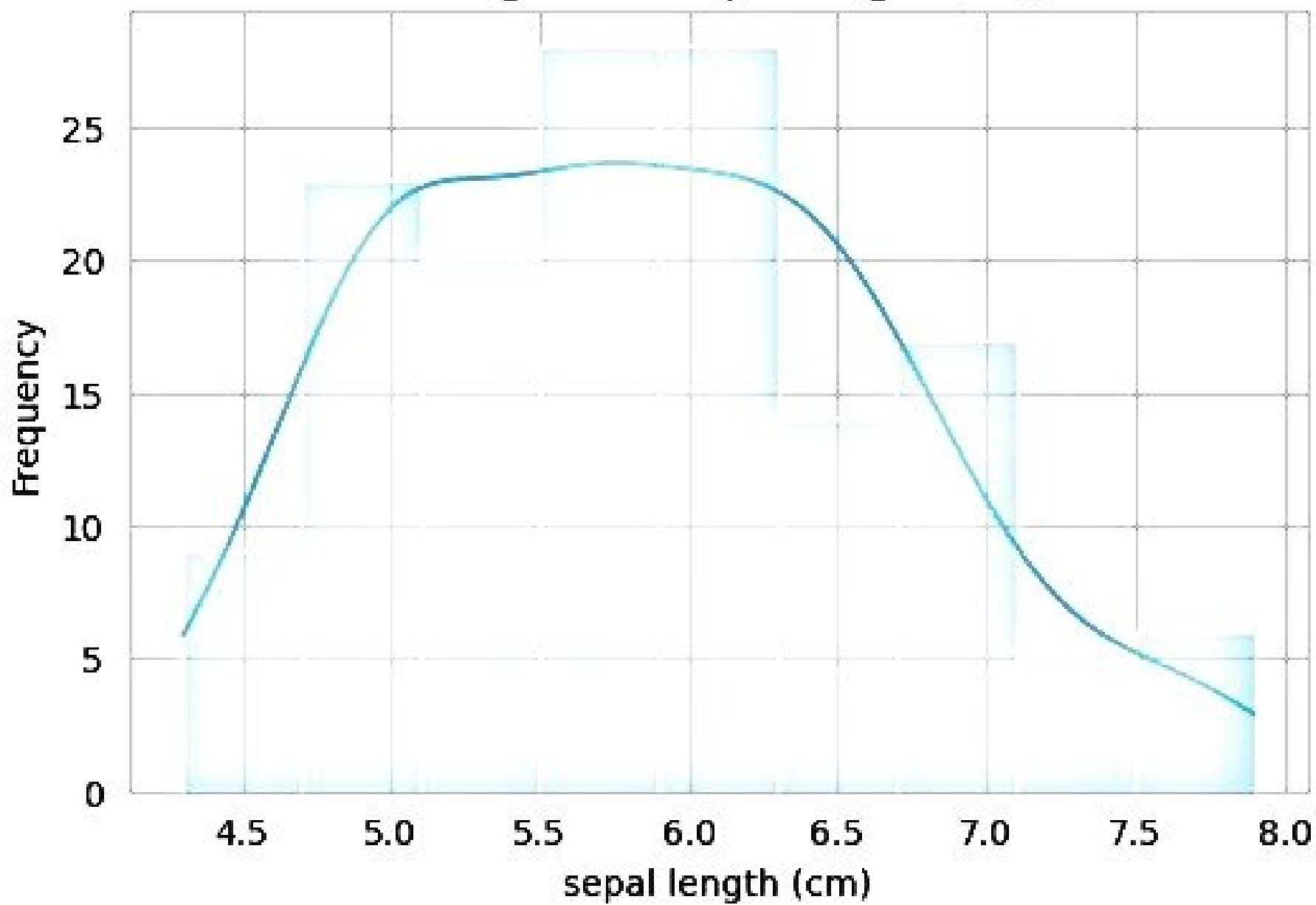
First 5 Rows:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

	species
0	setosa
1	setosa
2	setosa
3	setosa
4	setosa

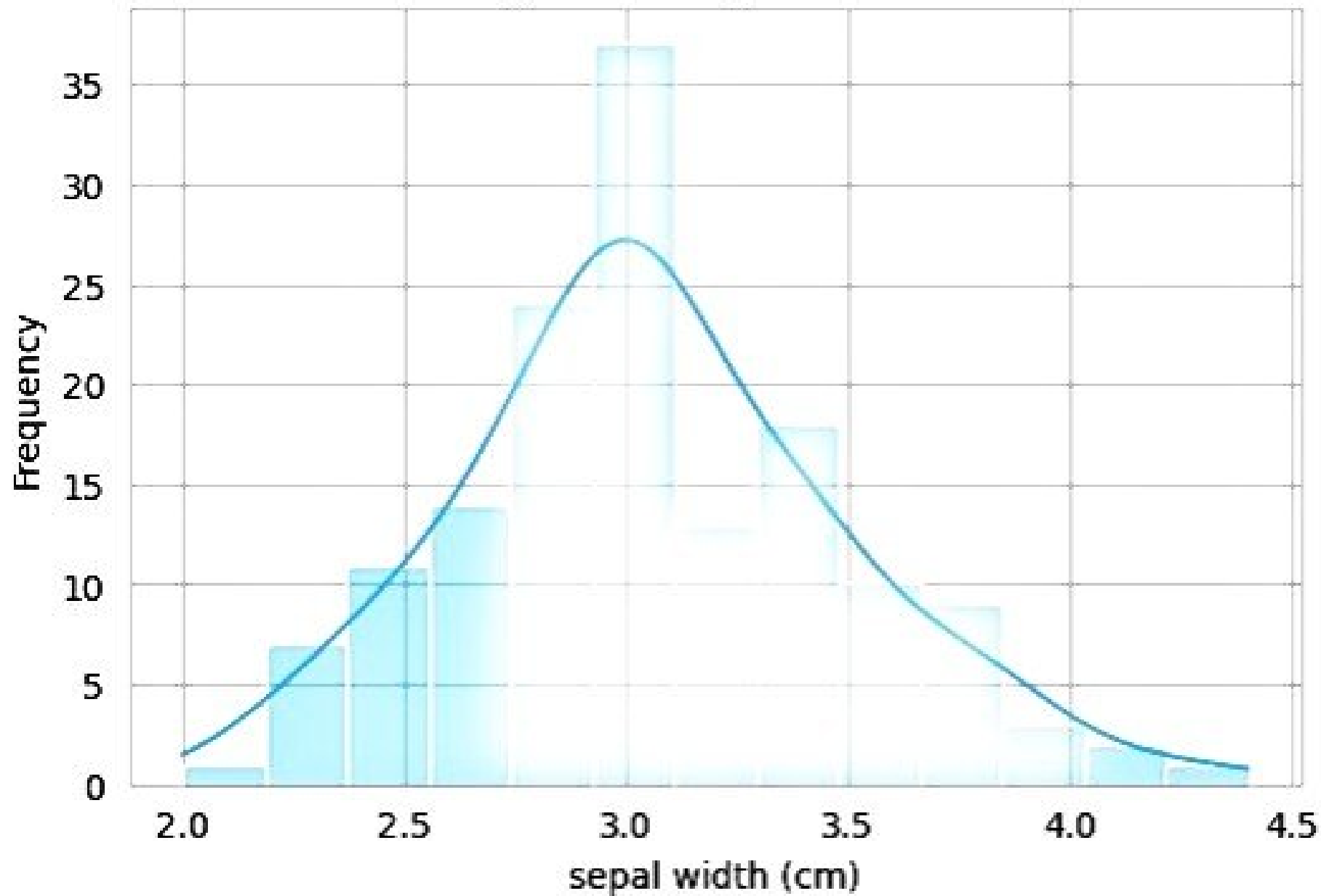
3 setosa
4 setosa

Histogram of sepal length (cm)



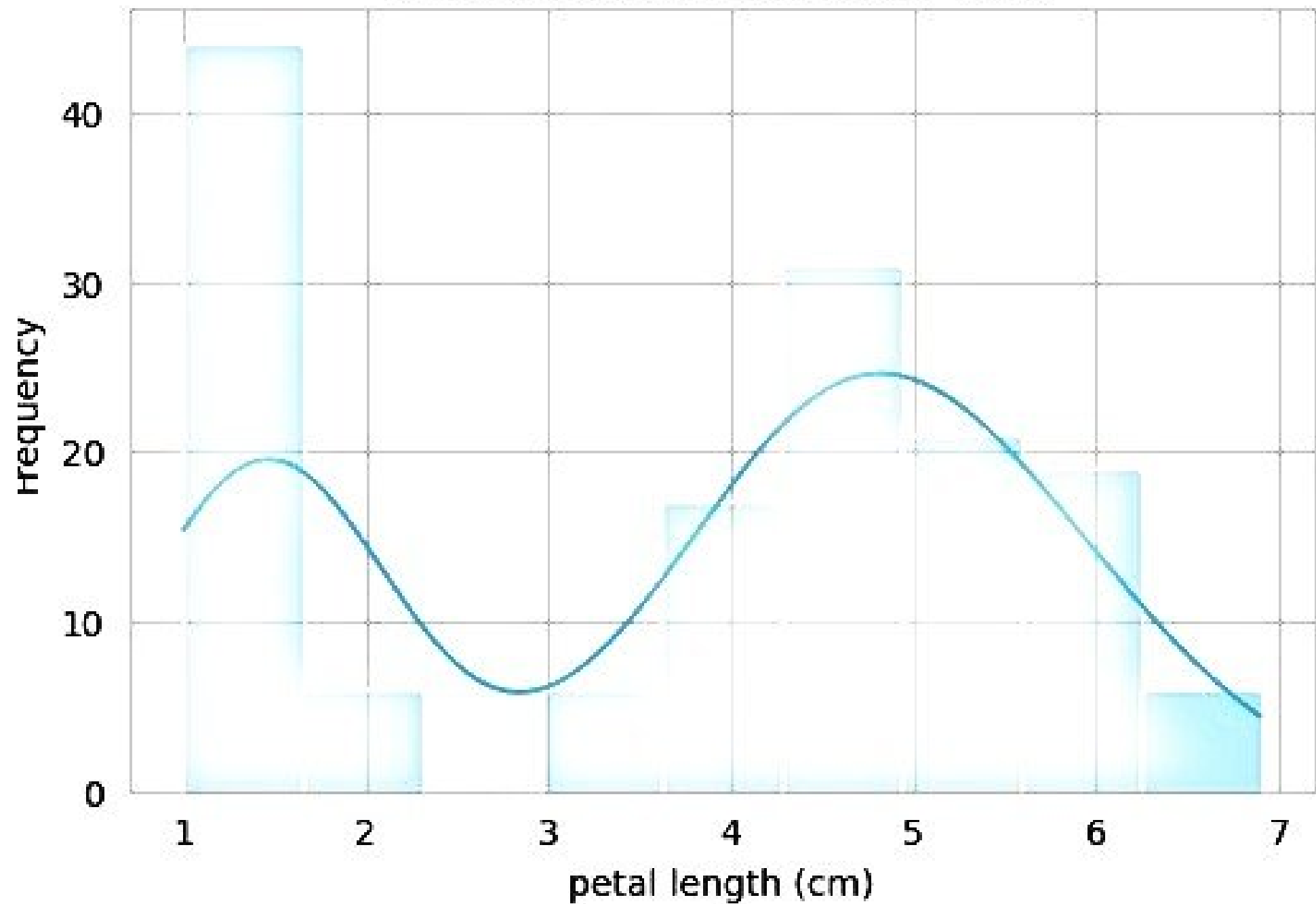
sepal length (cm)

Histogram of sepal width (cm)

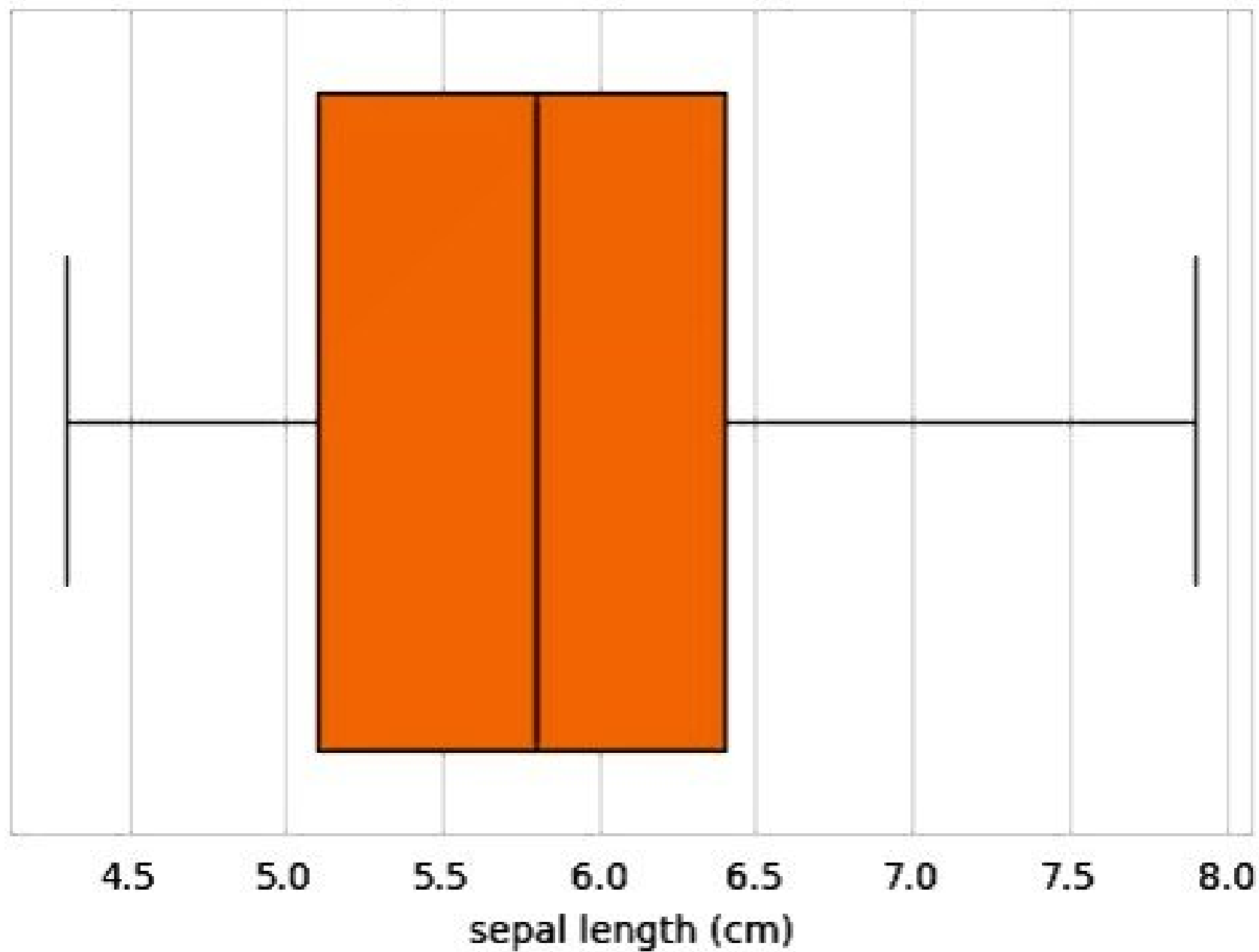


sepal width (cm)

Histogram of petal length (cm)

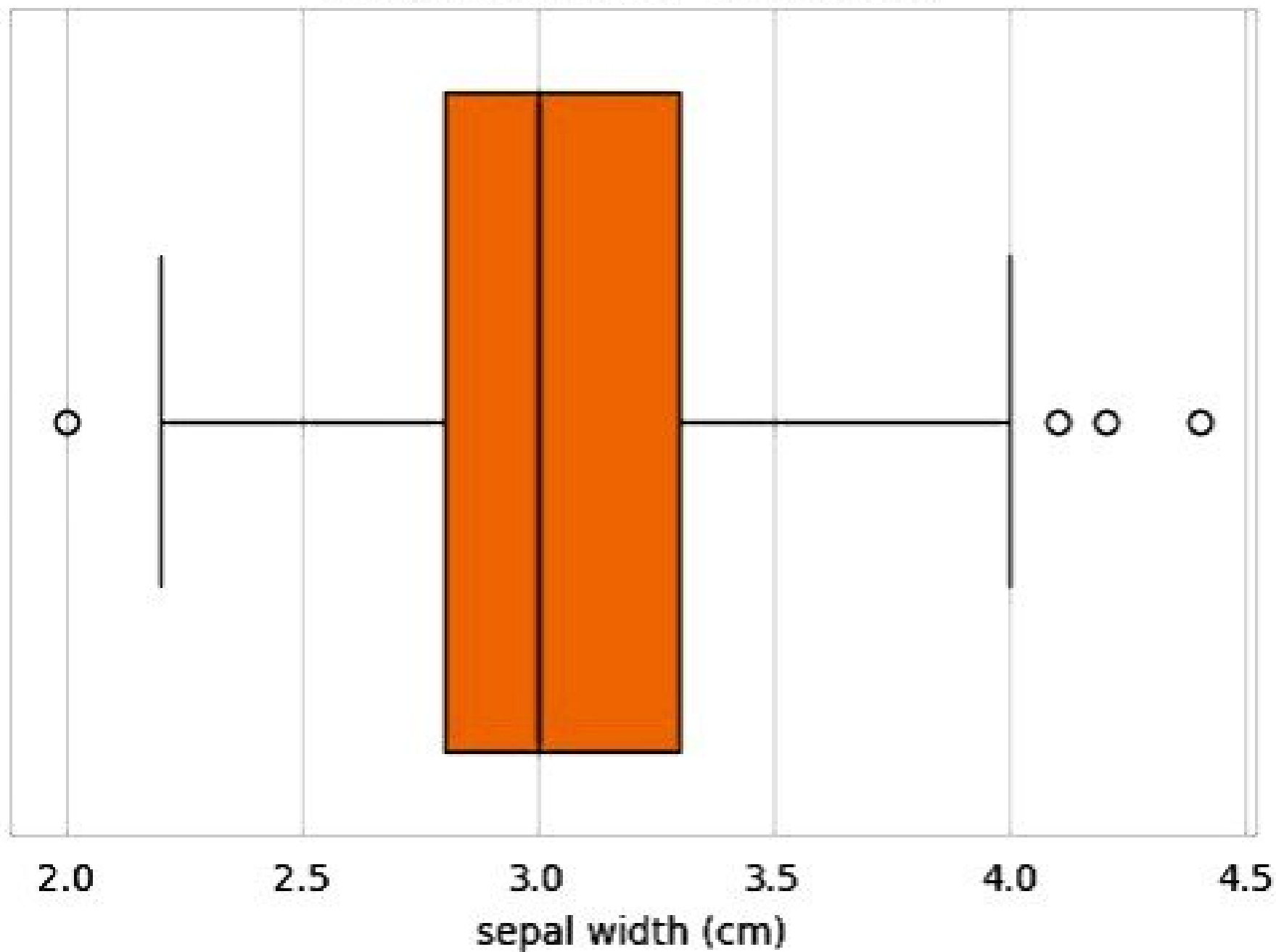


Boxplot of sepal length (cm)



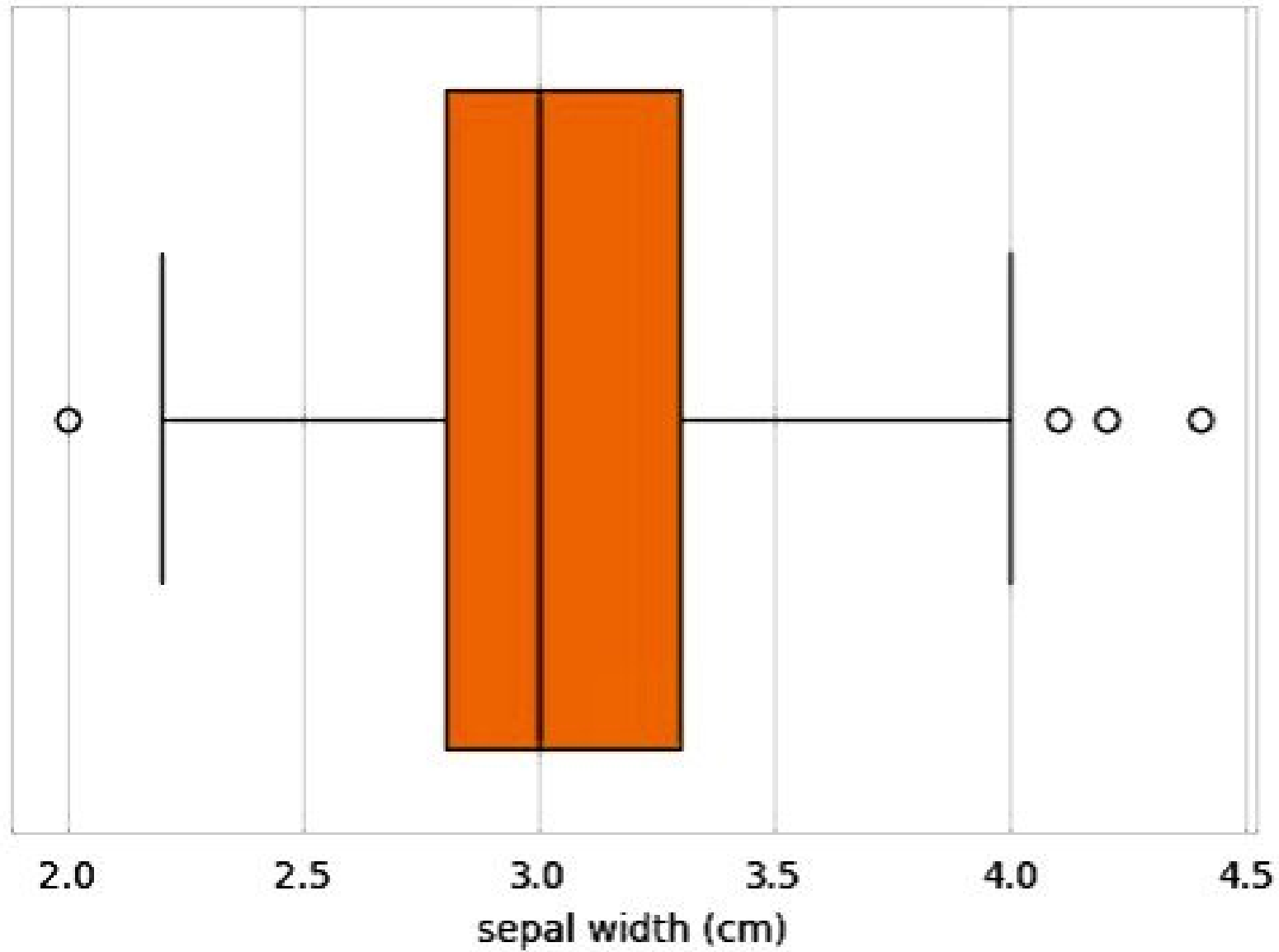
sepal length (cm)

Boxplot of sepal width (cm)

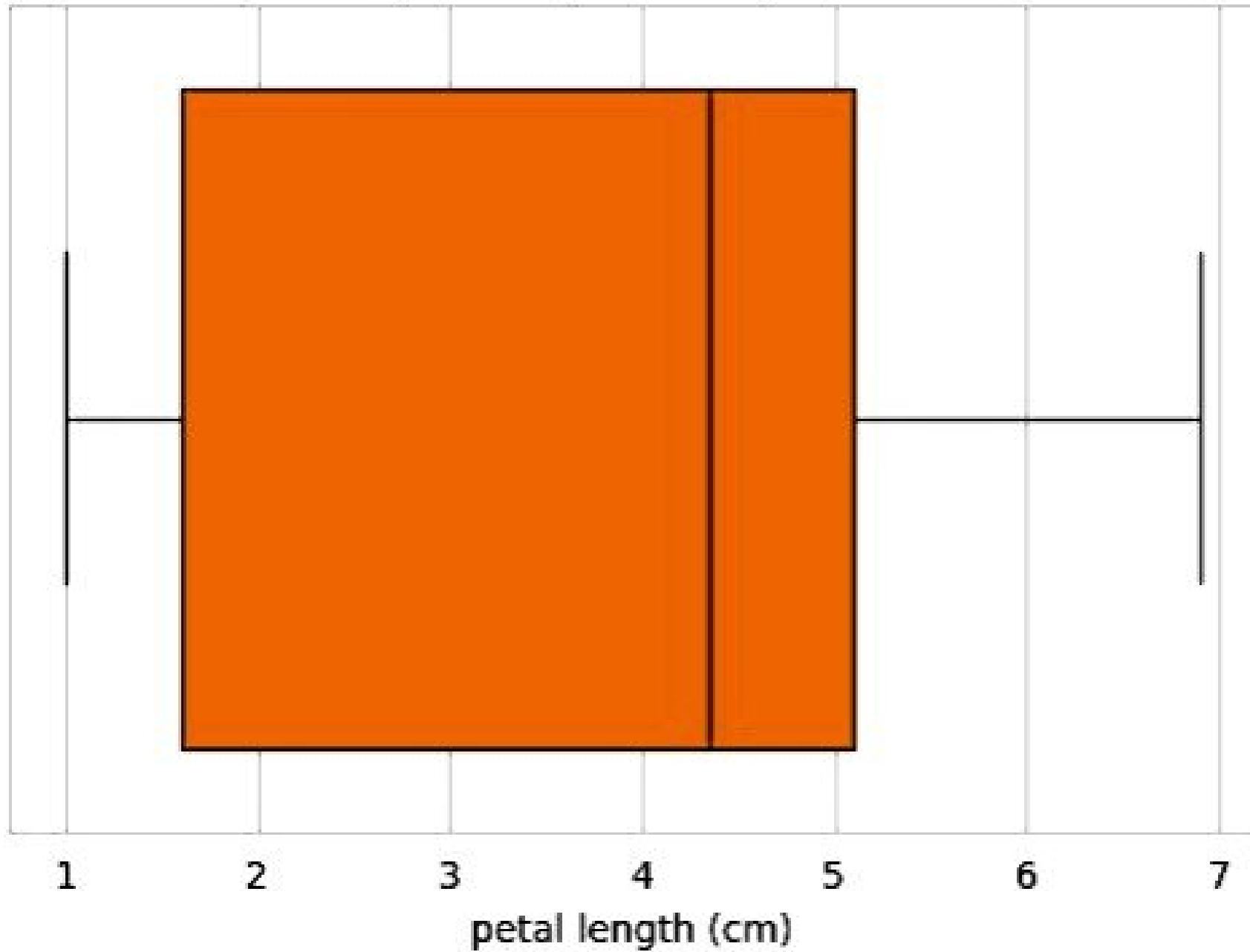


sepal length (cm)

Boxplot of sepal width (cm)

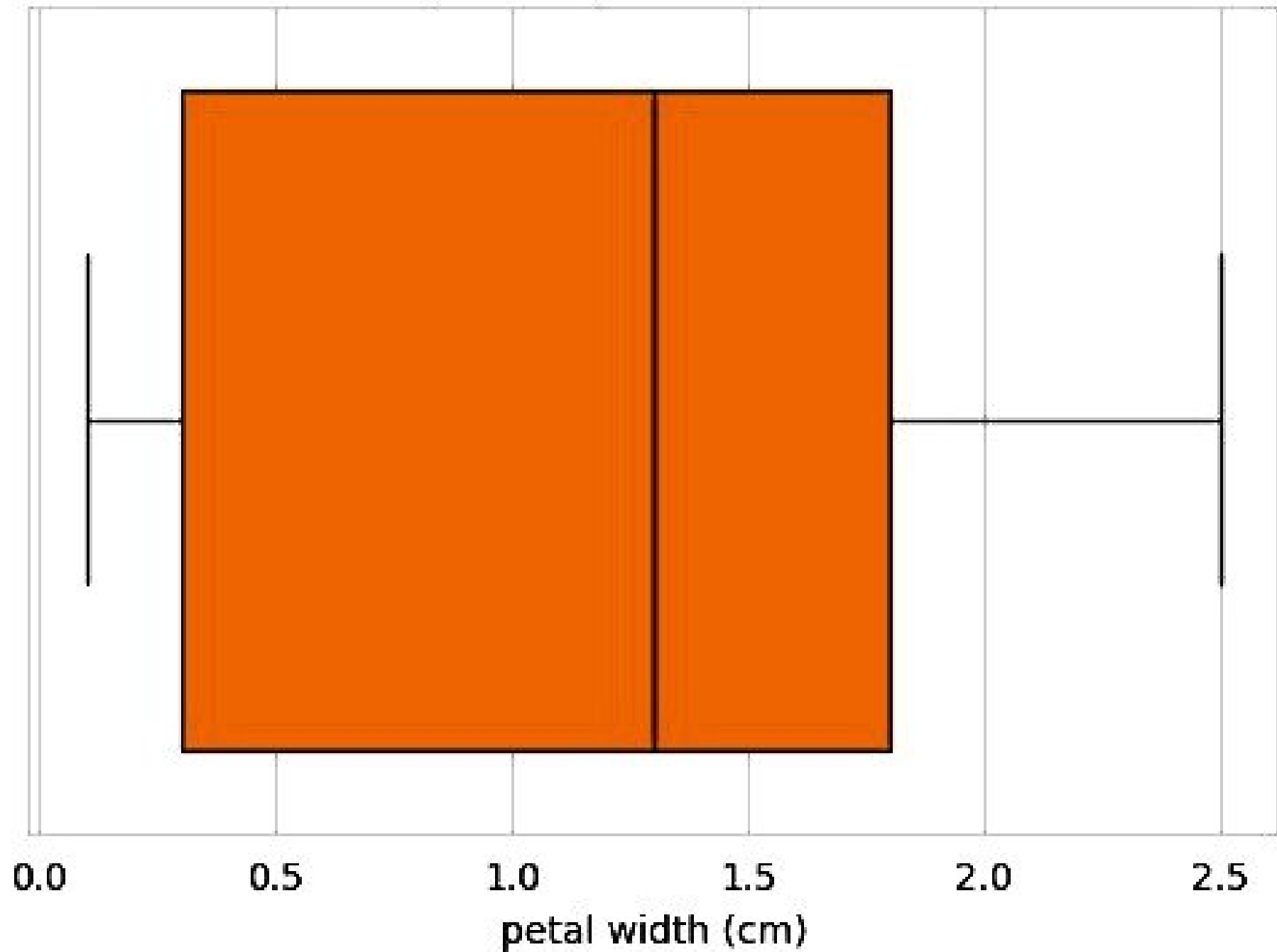


Boxplot of petal length (cm)

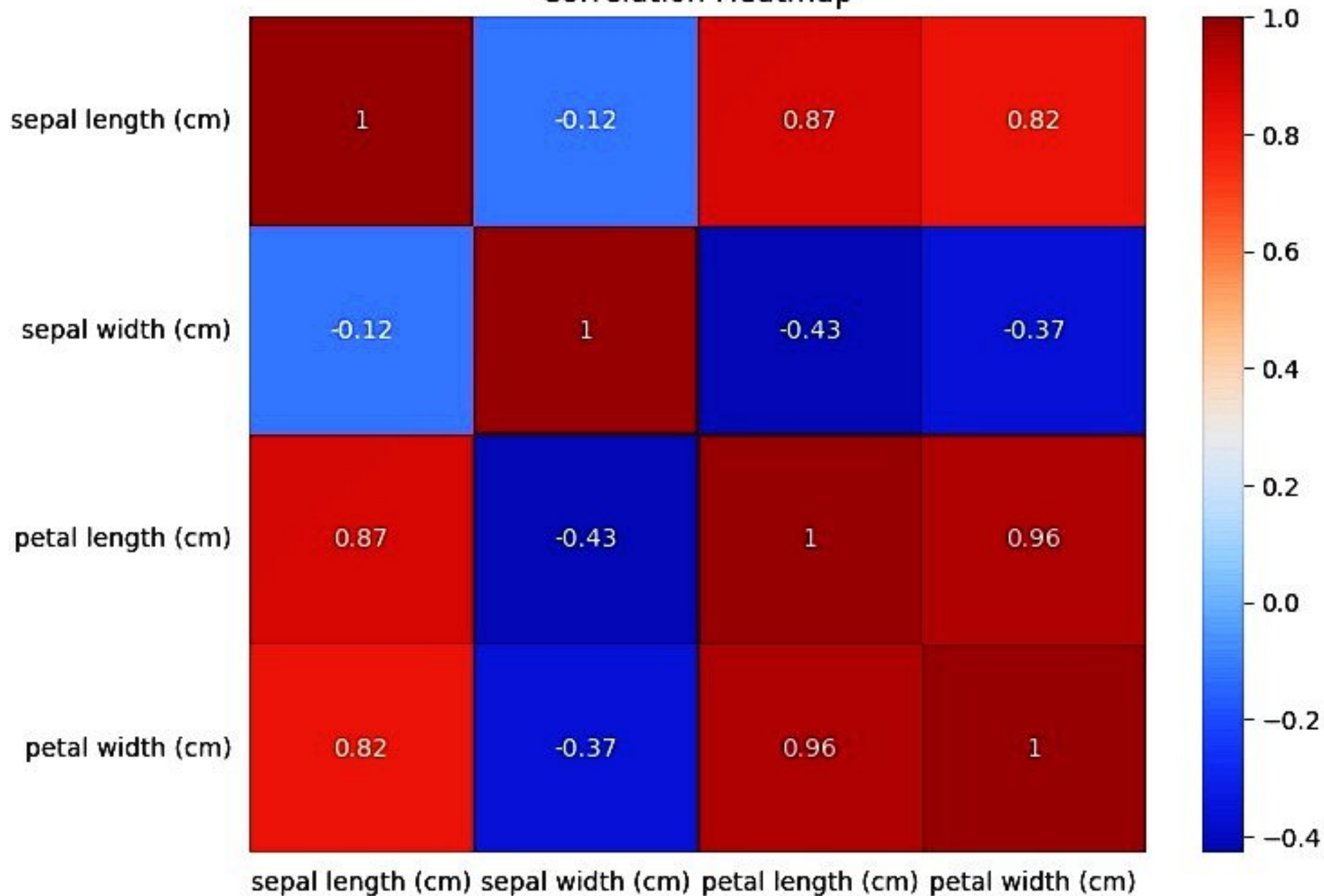


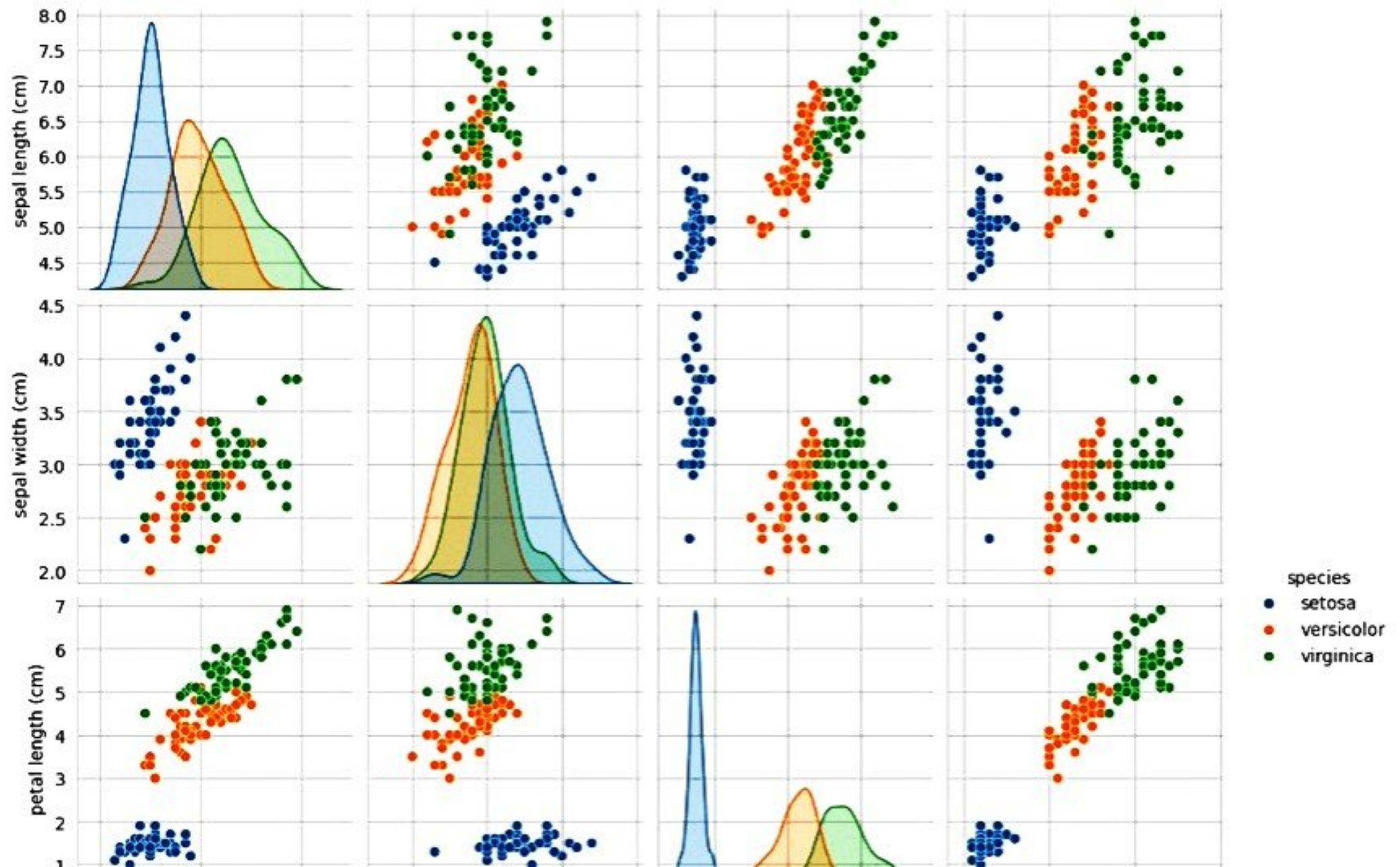
Boxplot of petal width (cm)

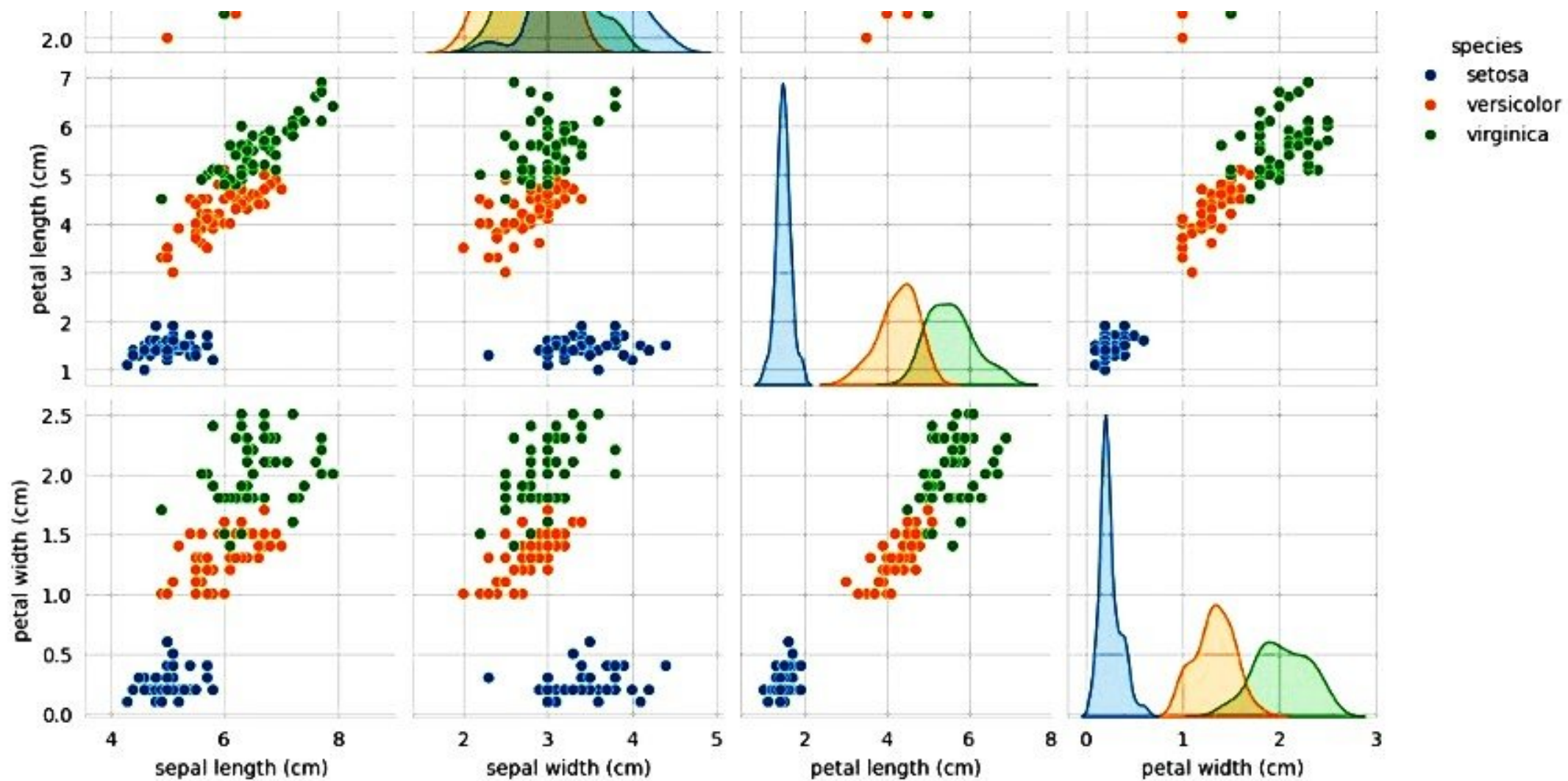
Boxplot of petal width (cm)

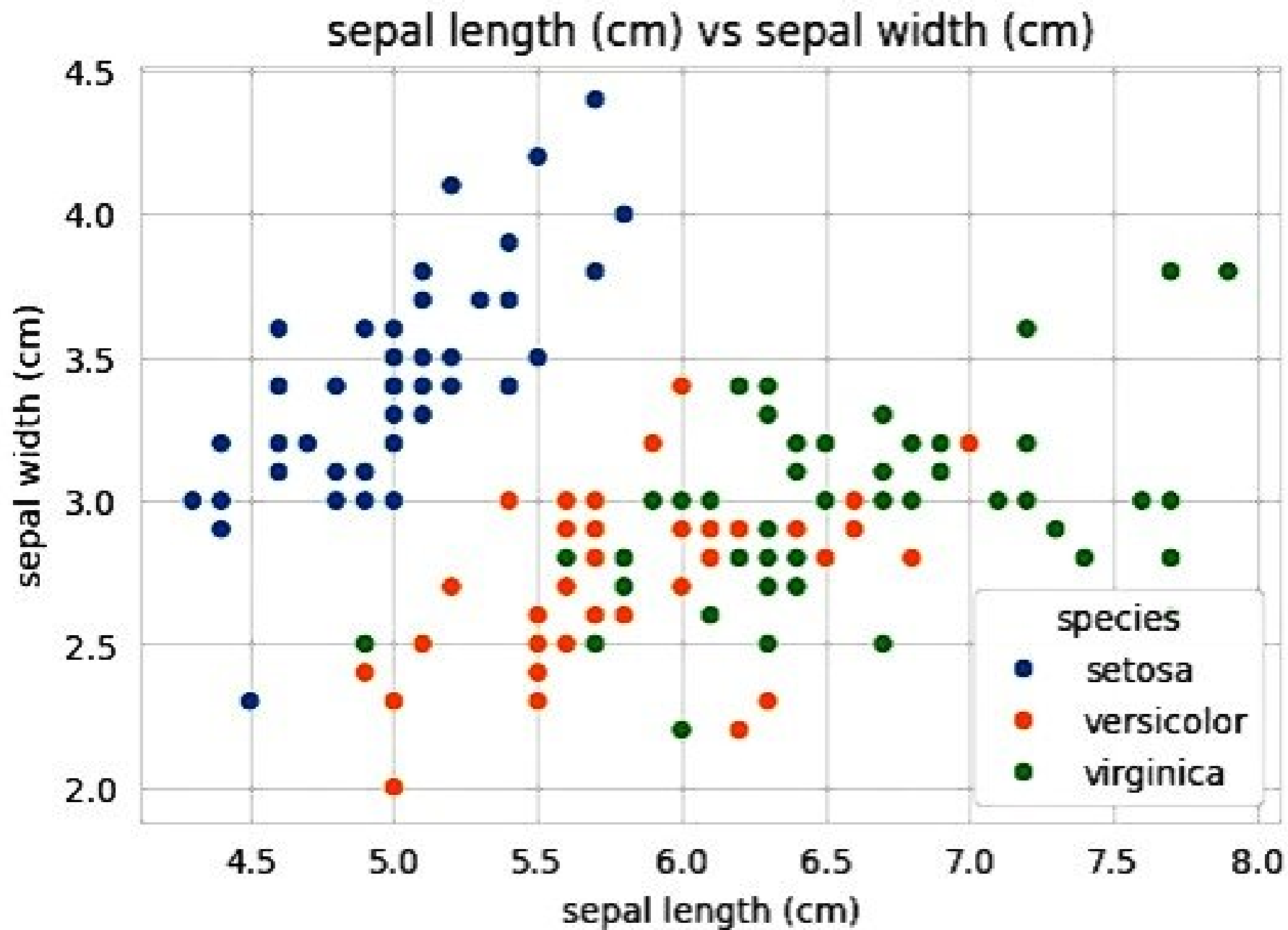


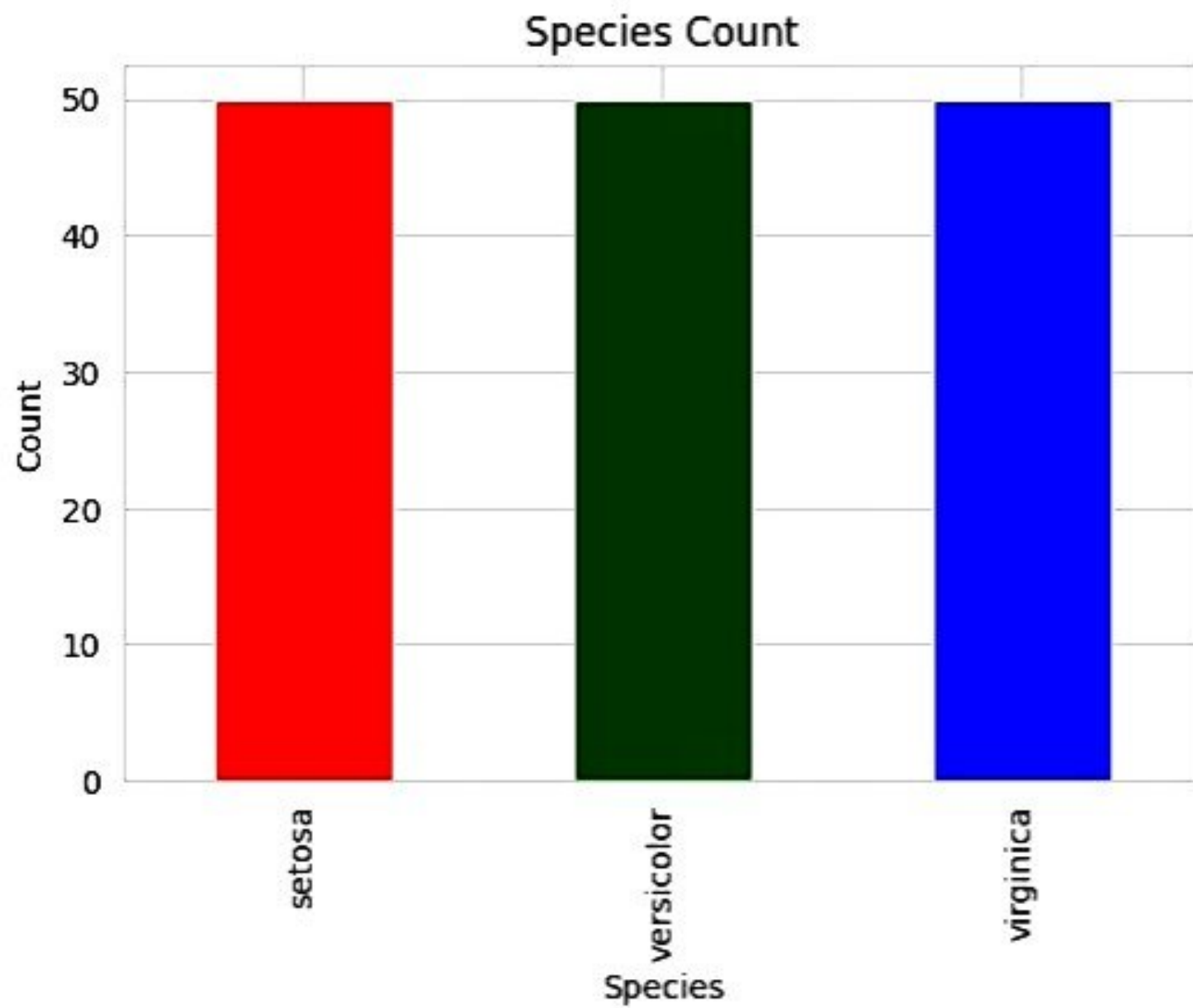
Correlation Heatmap



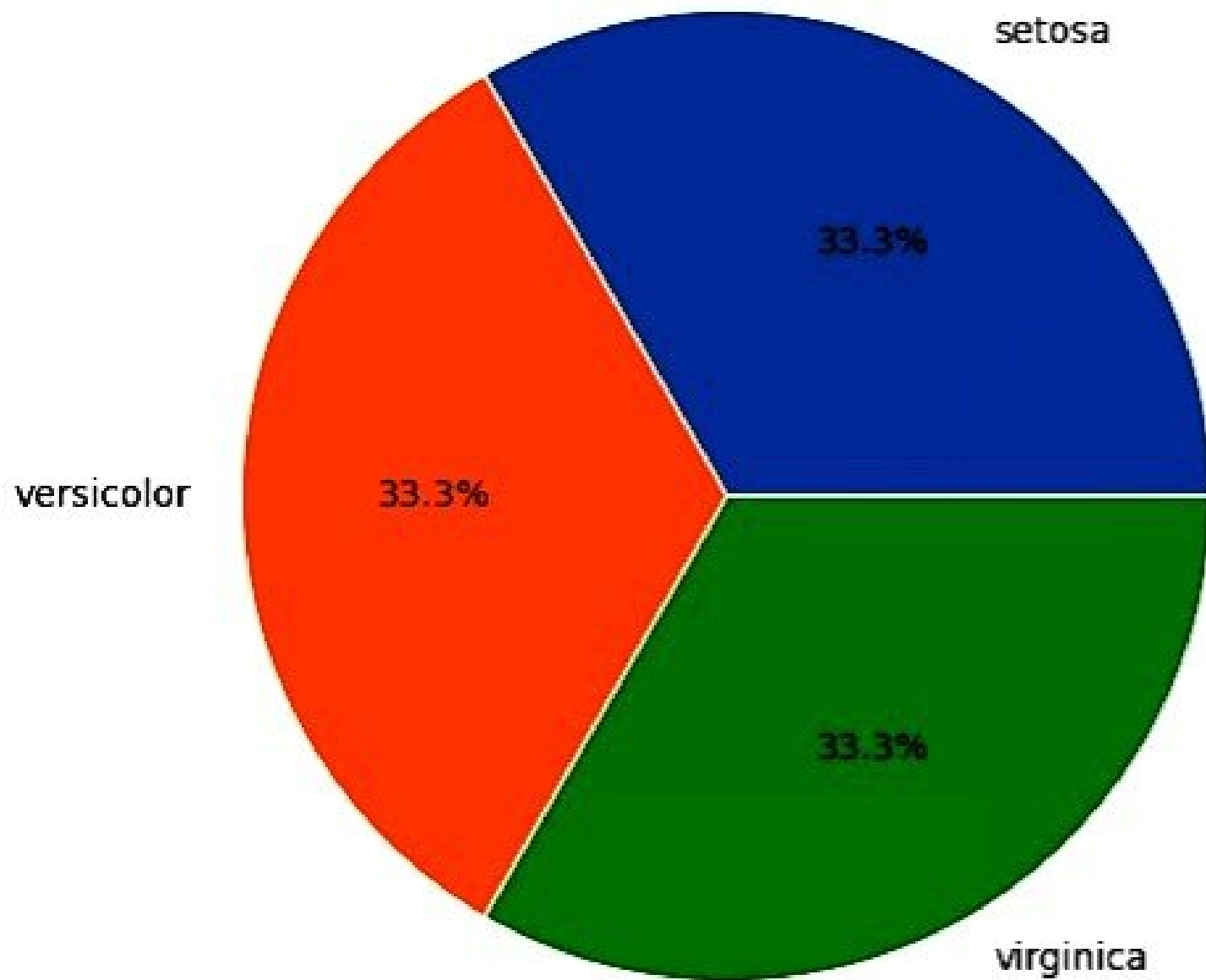




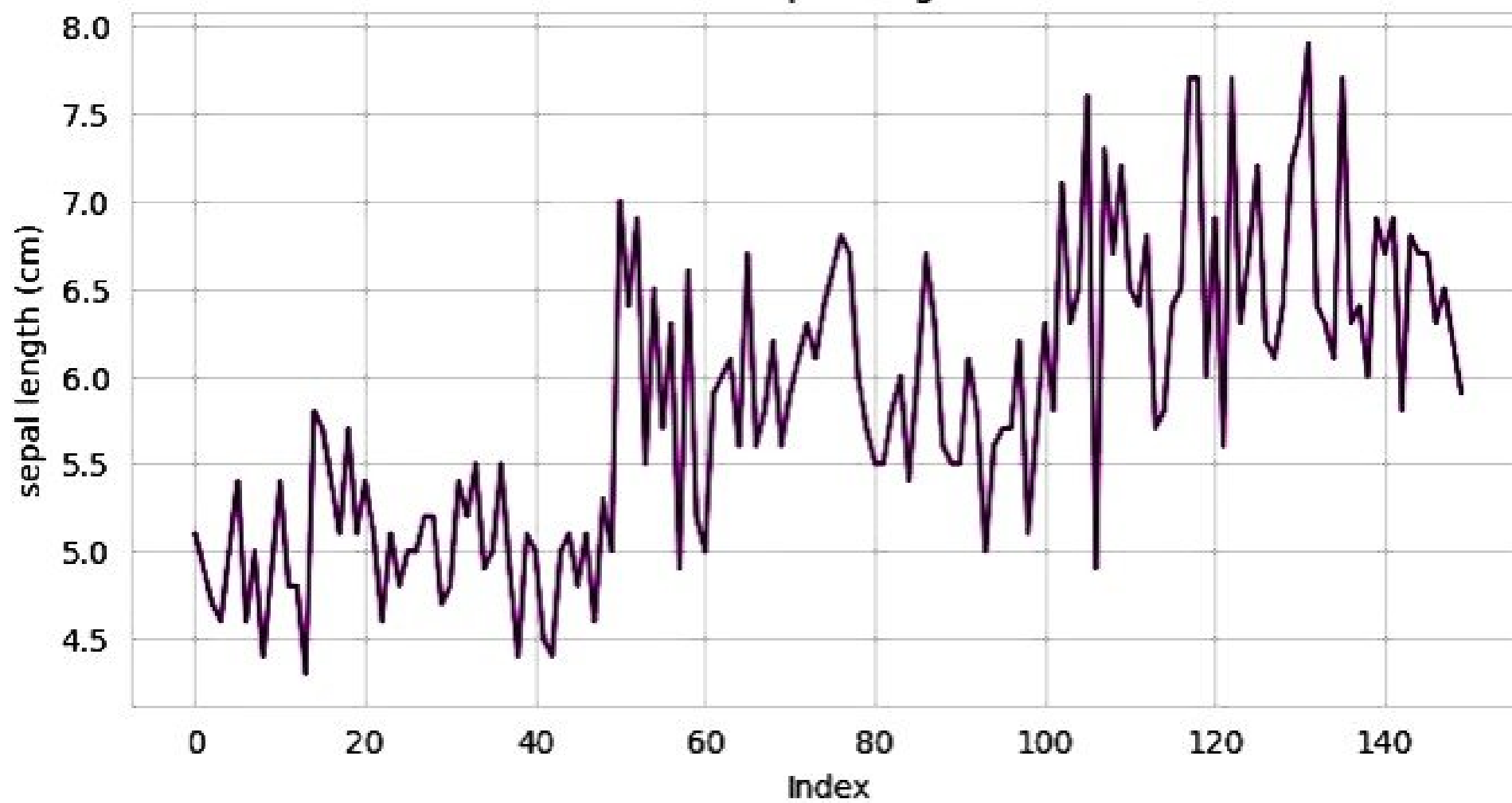




Species Distribution



Trend of sepal length (cm)



Data Visualization Completed Successfully!

Charts Generated:

- ✓ Histogram
- ✓ Box Plot
- ✓ Correlation Heatmap
- ✓ Pair Plot
- ✓ Scatter Plot
- ✓ Bar Chart
- ✓ Pie Chart
- ✓ Line Chart